

A Risk Management Framework for Intraday 0DTE Option Writing

Estimating Safe Strike Distances on NSE Expiry Days

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Motivation

On expiry days, 0DTE options undergo rapid theta decay while the remaining realized volatility of the underlying becomes increasingly predictable as the morning session progresses. The variance risk premium (VRP) — the systematic tendency for implied variance to exceed realized variance — provides the economic motivation for selling options on such days. However, this framework does not directly measure the VRP, as that would require a contemporaneous option chain to compare implied against realized variance. Instead, the approach is more precisely described as a realized variance forecasting framework that exploits the VRP indirectly: by forecasting remaining realized variance from morning price activity and the prevailing volatility regime, one can identify conditions under which the afternoon is likely to be quiet, and position short strikes at statistically grounded tail distances. The VRP enters empirically through the residual structure of the forecasting model — the systematic tendency of realized variance to come in below the model’s prediction confirms that the market prices in more variance than subsequently materializes, but this is a consequence of the framework rather than its direct input. The objective is not directional prediction but disciplined strike selection: answering the question of where it is statistically safe to write.

Setup

Let t denote any time in the entry window $[t_{\text{open}}, t_{\text{close}}]$ and let $T = 14:30$ denote the mandatory exit time. Define the remaining realized variance as

$$RV_{t \rightarrow T} = \sum_{s=t}^T r_s^2 \quad (1)$$

where $r_s = \log(P_s/P_{s-1})$ is the one-minute log return at bar s . The central research question is whether $RV_{t \rightarrow T}$ is forecastable from information available at t , and whether the option market’s implied variance over the same horizon systematically exceeds it.

Forecasting Remaining Realized Variance

Feature Selection via Gradient Boosting

Prior to specifying the parametric forecasting model, a richer candidate feature set was evaluated using a gradient boosted trees model (CatBoost) trained on expiry-day minute data. The candidate features included realized variance accumulated over multiple intraday lookback windows (15, 30, and 75 minutes, and the full morning session), a variance acceleration ratio, the cumulative intraday

return, the first-minute return, the maximum absolute one-minute return as a jump indicator, time-of-day features, the prior expiry day’s full-session realized variance, and the prior-day India VIX close. The model was evaluated by expanding-window walk-forward out-of-sample validation, and feature importances were computed on the full-sample fit.

Across both indices and all anchor times, two features consistently dominated: RV_{morn} , the morning realized variance from session open to the anchor, and VIX_{-1} , the prior trading day’s India VIX close. The former captures the intraday volatility regime as revealed by actual price activity; the latter provides a cross-day regime signal that remains informative even after conditioning on morning RV. All remaining features contributed negligibly to out-of-sample predictive accuracy. This motivates a parsimonious two-feature specification for the deployable model, which admits a closed-form log-linear structure and can be implemented without a live inference pipeline.

Two-Feature Log-Linear Model

The morning realized variance accumulated from session open to the anchor time $t^* = 11:00$ is

$$RV_{\text{morn}} = \sum_{s=09:16}^{11:00} r_s^2. \quad (2)$$

The prior trading day’s closing India VIX value, denoted VIX_{-1} , provides a regime-level signal available before market open and free of lookahead. The forecasting model is log-linear:

$$\log \widehat{RV}_{t^* \rightarrow T} = \alpha + \beta_1 \log RV_{\text{morn}} + \beta_2 \log VIX_{-1} \quad (3)$$

with coefficients $(\alpha, \beta_1, \beta_2)$ estimated separately for each index by ordinary least squares on expiry-day minute data. A one-feature fallback omitting VIX_{-1} is retained for sessions where the feed is unavailable. Model parameters are validated by expanding-window walk-forward out-of-sample evaluation.

Residual Distribution

Let $\varepsilon = \log RV_{t^* \rightarrow T} - \log \widehat{RV}_{t^* \rightarrow T}$ denote the out-of-sample prediction residual. The residuals are fit to a location-scale Student- t distribution:

$$\varepsilon \sim t(\nu, \mu_\varepsilon, \sigma_\varepsilon) \quad (4)$$

with degrees of freedom ν , location μ_ε , and scale σ_ε estimated by maximum likelihood. The negative location parameter $\mu_\varepsilon < 0$ across all anchor times reflects the VRP directly: the model systematically overpredicts remaining realized variance, meaning the realized path comes in below forecast on the typical expiry day. This systematic bias is the edge being harvested.

Empirical Results

CatBoost: Full Feature Set

Table 1 reports the walk-forward out-of-sample R^2 for the full 12-feature CatBoost model across anchor times, evaluated on expiry-day data from 2019–2025 (Nifty, 313 days) and 2016–2025 (BankNifty, 445 days). The initial training window used 60% of expiry days, with each subsequent day predicted using an expanding window fit on all prior observations.

Predictability increases materially from 09:30 to 11:00, supporting the hypothesis that remaining realized variance becomes more forecastable as opening auction information is absorbed into price. The preferred entry time of $t^* = 11:00$ coincides with peak predictability across both indices.

Table 1: CatBoost walk-forward OOS R^2 (12-feature model, log RV space)

Anchor	Nifty	BankNifty
09:30	0.182	0.395
10:00	0.256	0.417
10:30	0.319	0.447
11:00	0.417	0.458
11:30	0.365	0.404
12:00	0.427	0.450

OLS Two-Feature Model

The VIX lift — the incremental OOS R^2 from adding the prior-day VIX to the morning realized variance — is +0.148 for Nifty and +0.067 for BankNifty, confirming that the regime signal is genuinely additive beyond morning realized variance alone.

Table 2: OLS walk-forward OOS performance at anchor $t^* = 11:00$

Model	Nifty		BankNifty	
	OOS R^2	RMSE	OOS R^2	RMSE
2-feature (RV + VIX)	0.394	0.592	0.498	0.543
1-feature (RV only)	0.246	—	0.431	—
VIX lift	+0.148		+0.067	

Residual Distribution and Variance Risk Premium

Table 3 reports the fitted Student- t parameters on the OOS residuals from the two-feature model. The negative location parameter $\mu_\varepsilon < 0$ in both cases confirms the direction of the VRP: on a typical expiry day, realized remaining variance comes in below the model’s prediction, meaning the market prices in more variance than subsequently materializes. The degrees of freedom $\nu \approx 7$ for Nifty versus $\nu \approx 15$ for BankNifty indicate heavier residual tails in the more concentrated index, motivating the Student- t specification over a normal approximation in both cases.

Table 3: Fitted Student- t residual parameters at $t^* = 11:00$

Index	ν	μ_ε	σ_ε
Nifty	7.42	−0.179	0.467
BankNifty	15.26	−0.176	0.468

Diagnostic Plots

Figures 1 and 2 summarize the two-feature model diagnostics for BankNifty and Nifty respectively. Each figure shows, from top to bottom: the log RV scatter colored by prior-day VIX tercile (illustrating the regime dependence motivating the VIX feature), the out-of-sample walk-forward fit of the 2-feature versus 1-feature model against the actual remaining log RV, and the fitted

Student- t distribution against the empirical OOS residual histogram, overlaid with a normal fit for comparison.

Strike Selection

Conditional Quantile

For a desired tail probability α_q , the conditional quantile of the t -residual is

$$q_{\alpha_q} = F_t^{-1}(\alpha_q; \nu, \mu_\varepsilon, \sigma_\varepsilon) \quad (5)$$

where F_t^{-1} denotes the quantile function of the fitted t -distribution. The predicted log realized variance at this quantile is

$$\log \widehat{RV}^{(q)} = \log \widehat{RV}_{t^* \rightarrow T} + q_{\alpha_q}. \quad (6)$$

The corresponding return magnitude — the predicted price move at tail probability α_q — is

$$\hat{r}^{(q)} = \sqrt{\exp\left(\log \widehat{RV}^{(q)}\right)}. \quad (7)$$

The put strike K_P and call strike K_C of the short strangle are then

$$K_P = S_{t^*} \cdot \left(1 - \hat{r}^{(q_{\text{put}})}\right) \quad (8)$$

$$K_C = S_{t^*} \cdot \left(1 + \hat{r}^{(q_{\text{call}})}\right) \quad (9)$$

where S_{t^*} is the spot price at the anchor time, rounded to the nearest tradeable strike increment.

Safety Floor

Model uncertainty and the finite sample of historical expiry days motivate a hard lower bound on strike distance. A minimum percentage $\delta_{\min}$ is imposed such that

$$\hat{r}^{(q_{\text{put}})} \geq \delta_{\min}, \quad \hat{r}^{(q_{\text{call}})} \geq \delta_{\min}. \quad (10)$$

When the model's predicted quantile is tighter than $\delta_{\min}$ — as occurs on anomalously quiet mornings — the floor overrides the model output. This activation is reported explicitly in the trading indicator, so the practitioner always knows whether the model or the floor is setting the strike.

Operational Framework

The strategy operates exclusively on expiry days within the entry window [10:30, 12:00] with a mandatory exit at $T = 14:30$. At the anchor time $t^* = 11:00$, the morning realized variance is computed from the intraday one-minute price series, the prior-day VIX close is fetched, and the model produces a predicted remaining variance and associated conditional strike quantiles for the short strangle. The practitioner applies a discretionary event filter — excluding scheduled macro events, central bank announcements, and budget days — before entering. Position sizing and intraday stop-loss decisions remain discretionary, with the quantitative model serving strictly as the strike-selection anchor.

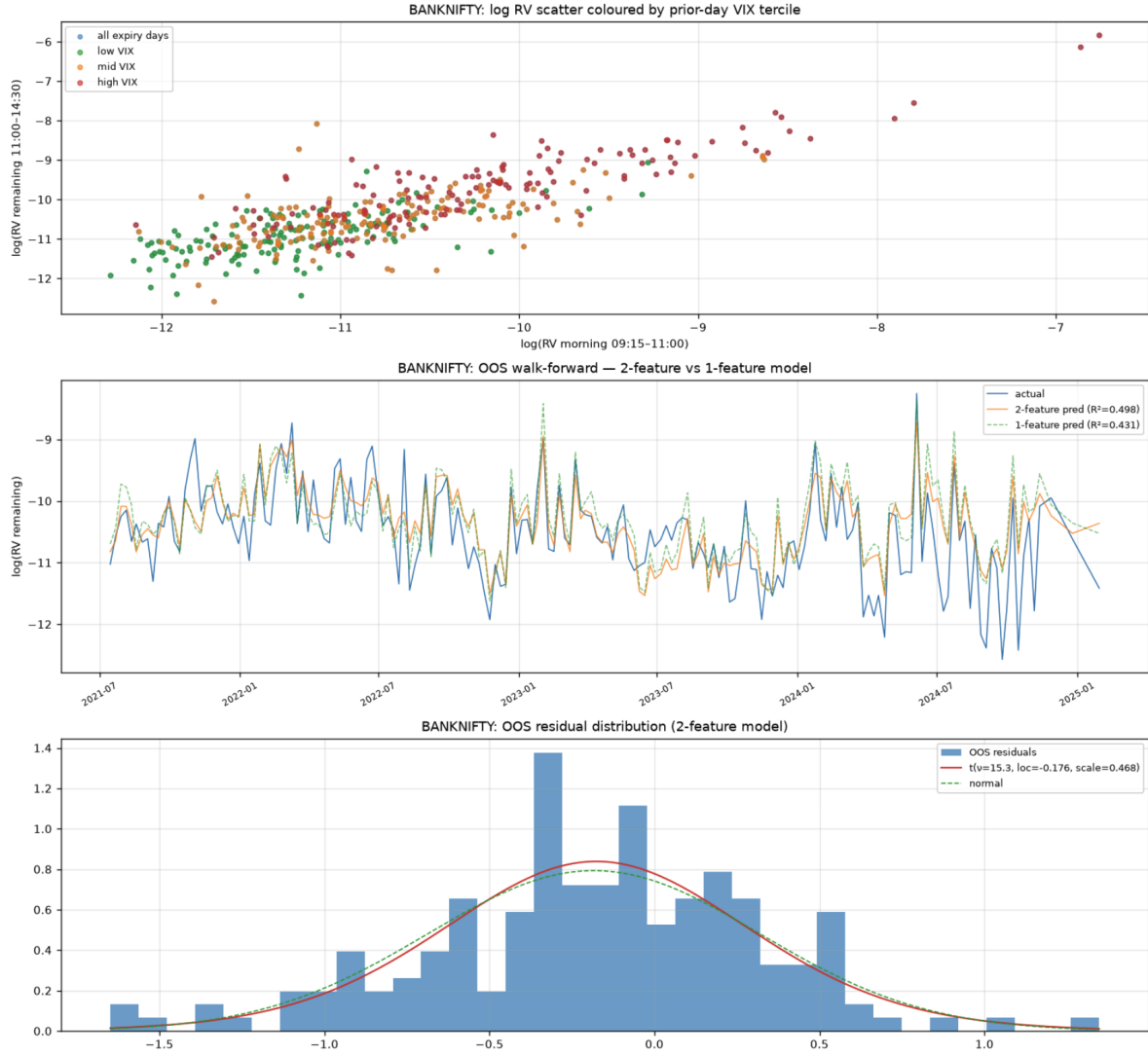


Figure 1: BankNifty: VIX-tercile scatter, OOS walk-forward fit, and residual distribution for the two-feature model.

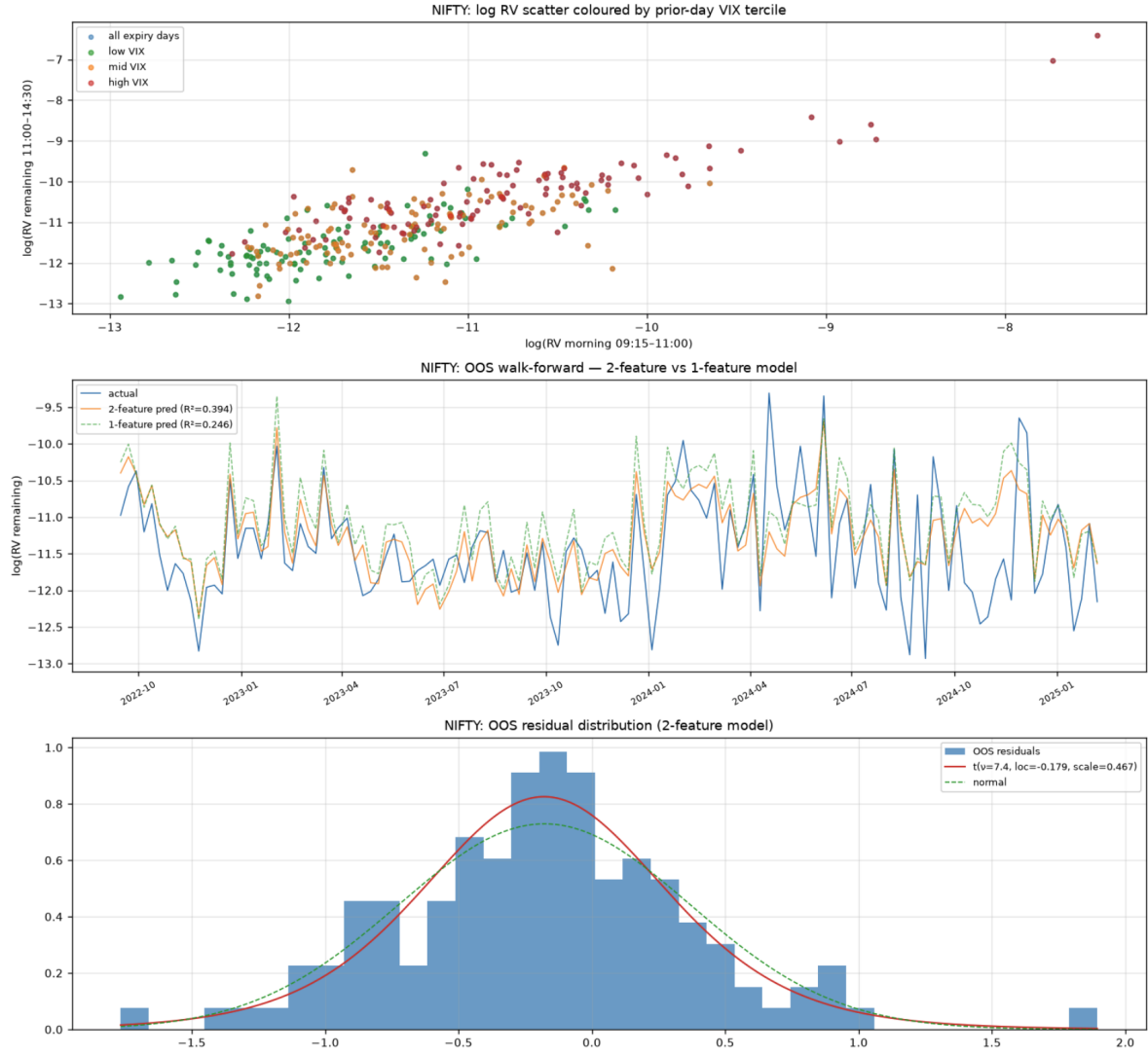


Figure 2: Nifty: VIX-tercile scatter, OOS walk-forward fit, and residual distribution for the two-feature model.